

TECHNICAL REPORT

Surface Water Quality

Complex hydrology and variability of nitrogen sources in a karst watershed

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Abstract

Streams draining karst areas with rapid groundwater transit times may respond relatively quickly to nitrogen reduction strategies, but the complex hydrologic network of interconnected sinkholes and springs is challenging for determining the placement and effectiveness of management practices. This study aims to inform nitrogen reduction strategies in a representative agricultural karst setting of the Chesapeake Bay watershed (Fishing Creek watershed, Pennsylvania) with known elevated nitrate contamination and a previous documented groundwater residence time of less than a decade. During baseflow conditions, streamflow did not increase with drainage area. Headwaters and the main stem lost substantial flow to sinkholes until eventually discharging along large springs downstream. Seasonal hydrologic conditions shift the flow and nitrogen load spatially among losing and gaining stream sections. A compilation of nitrogen source inputs with the geochemistry and the pattern of enrichment of $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ suggest that the nitrogen in streams and springs during baseflow represents a mixture of manure, fertilizer, and wastewater sources with low potential for denitrification. The pH and calcite saturation index increased along generalized flow paths from headwaters to springs and indicate shorter groundwater residence times in baseflow during the spring versus summer. Given the substantial investment in management practices, fixed monitoring sites could incorporate synoptic water sampling to properly monitor long-term progress and help inform management actions in karst watersheds. Although karst watersheds have the potential to respond to nitrogen reduction strategies due to shorter groundwater residence times, high nitrogen inputs, effectiveness of conservation practices, and release of legacy nutrients within the karst cavities could confound progress of water quality goals.

Abbreviation: USGS, U.S. Geological Survey.

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Plain Language Summary

Water and excess nitrogen in a karst landscape can travel rapidly underground through a complex network of dissolved pathways in the carbonate rock (e.g., limestone) before departing to streams and springs. To inform management and monitoring efforts of nitrogen, we used near simultaneous measurements (synoptic sampling) of water quality across multiple stream and spring locations during seasonal high and low stream baseflow. This study revealed that the main sources of nitrogen include manure, fertilizer, and wastewater with a low potential for denitrification (conversion of nitrate to nitrogen gas). Depending on the season or location in the watershed, the nitrogen load shifted among losing and gaining stream sections. Establishing monitoring at fixed locations without synoptic sampling could be problematic for assessing management progress. Karst areas have the potential to reduce nitrogen rapidly, but inputs, conservation effectiveness, and legacy nutrients confound progress.

1 | INTRODUCTION

Estuarine waters around the globe have experienced declines in water quality and impairments to aquatic ecology resulting from increased human population, sprawling development, and intensive agriculture (Rabalais et al., 2009). For example, over the past century, excess nitrogen loading has taken a toll on the ecological and economic aspects of the Chesapeake Bay, driven by changes in climate, hydrology, atmospheric deposition, land use, and agricultural and urban management (Clune et al., 2021). Nitrogen from atmospheric deposition and wastewater increased substantially from 1950 but has decreased in recent decades following the passing of the Clean Air Act and the implementation of enhanced N removal technologies (Ator, 2019; Chesapeake Bay Program, 2017; Linker et al., 2013; Lloret & Valiela, 2016). Presently, the greatest opportunities for future nitrogen reductions in the Chesapeake Bay watershed are in agricultural areas, where the intensification of crop and animal production has seen widespread use of chemical fertilizers and manure (Clune et al., 2021).

Nearly half of the nitrogen load to streams in the Chesapeake Bay watershed is delivered as nitrate from groundwater that discharges as baseflow (Bachman et al., 1998; Phillips et al., 1999). Additionally, the highest baseflow nitrate concentrations are often associated with intensive agricultural land use and carbonate geology (Ator et al., 2011; Clune, Crawford, Chappell, & Boyer, 2020; Wherry et al., 2021). As carbonate rock chemically dissolves, it forms karst terrains (losing streams, sinkholes, underground caves, and springs) that are highly susceptible to contamination as the short residence times of pollutants, such as nitrate, limit the potential for denitrification (Hartmann et al., 2014; Husic et al., 2021). Focused management efforts in agricultural carbonate settings with rapid transit times show evidence of being

an effective approach toward reducing nitrogen loads to the Chesapeake Bay (Ator et al., 2020).

Although the association of elevated nitrate concentrations, agricultural land use, and carbonate geology can be indicative of nitrogen sources, more detailed studies that better account for the spatial heterogeneity and temporal variability in sources, biogeochemical processes, and residence time could support resource managers in their decision-making processes (Böhlke & Denver, 1995; Canion et al., 2020; Chang et al., 2022; Clune & Denver, 2012; Heffernan et al., 2012; Husic et al., 2019; Hyer et al., 2016; Jones & Smart, 2005; Katz, 2004; Katz et al., 1999; Musgrove et al., 2016; Puig et al., 2013; Yue et al., 2020). For instance, the ratios of nitrate isotopes ($\delta^{15}\text{N}$ and $\delta^{18}\text{O}$) have been successfully used in watersheds to help determine distinct nitrate signatures (atmospheric, manure, fertilizer, sewage, etc.) and provide evidence of fractionation processes such as denitrification (Clune & Denver, 2012; Granger & Wankel, 2016; Kendall et al., 2007; Xue et al., 2009).

Nitrate from multiple sources contaminate groundwater supplies in karst settings of the Chesapeake Bay watershed (Greene et al., 2005). Local assessments and water-quality monitoring reports for small carbonate watersheds such as Fishing Creek in Pennsylvania have documented degradation of the stream water quality and groundwater used for drinking water supplies (CDS, 2004; Clune & Cravotta, 2020; RETTEW, 2004). The presence of nitrate in groundwater was often cooccurring with *Escherichia coli*, which was commonly found in domestic wells that are unsuitable for drinking without treatment, prompting sampling of surface water for possible sources (Clune & Cravotta, 2020). Current nutrient reduction milestones for local and regional water quality goals (e.g., total maximum daily loads) in small watersheds and the greater Chesapeake Bay watershed have necessitated rapid

small-scale multidisciplinary approaches that can provide the relative contributions of nitrogen sources for conservation planning.

To inform nitrogen reduction strategies in karst regions, we investigated the occurrence and distribution, biogeochemical processes, potential sources, and residence time of nitrate in a representative agricultural carbonate setting of the Chesapeake Bay watershed (Fishing Creek watershed, Pennsylvania) (Figure 1). To meet these objectives, stream and spring water chemistry data were collected in the Fishing Creek watershed in 2019 during seasonal high (April) and low (August) baseflow when groundwater was the dominant contribution of streamflow. The study also aimed to incorporate groundwater data collected in June 2017 from domestic wells in the watershed (Clune & Cravotta, 2020) within the range of previous estimates of the residence time (Focazio et al., 1998; Lindsey et al., 2003) for comparability with spring and surface water collected during baseflow conditions.

2 | MATERIALS AND METHODS

2.1 | Study area and hydrogeological setting

The Fishing Creek watershed (157.2 km²) is located in central Pennsylvania (Figure 1). The study area is in the upper portion of Fishing Creek watershed that is part of the Susquehanna River basin that drains eventually to the Chesapeake Bay. In this karst setting of the Valley and Ridge physiographic province (Fenneman & Johnson, 1946), groundwater provides the largest contribution of flow in streams and is the dominant source of dissolved solids, including nitrate (Jacobson & Langmuir, 1970; Langmuir, 1971; Shuster & White, 1972; Taylor et al., 1983). The headwater streams along the valley sides originate on forested slopes underlain by siliciclastic rocks that often sink into carbonate bedrock along the valley floor that is composed of predominately limestone and dolomite (Horton et al., 2017; Taylor et al., 1983). Numerous sinkholes allow for rapid transport of surface contaminants into the subsurface aquifer (Harmon et al., 1975). Most of the mountain tributaries drain into sinkholes before reaching Fishing Creek and the main channel consists of alternating losing and gaining reaches that eventually pass through a narrow mountain gap (Figures 1 and 2). The second (S5) and fourth (S4) largest springs in Pennsylvania discharge into the lower portion of the drainage area (Flippo, 1974). The “narrows” section just downstream of the study area is designated a high-quality cold-water fishery and is a renowned trout stream by anglers (PFBC, 2010).

The local climate provides for a wet season of increased precipitation from April to June, with a mean annual rainfall of 110–114 cm/year, of which approximately 51–57 cm recharges the aquifer (Risser et al., 2008). The topography

Core Ideas

- Complex hydrology in agricultural carbonate watersheds challenges effective conservation practice implementation.
- Primary sources of nitrogen include manure, fertilizer, and wastewater sources with low potential for denitrification.
- Hydrologic conditions shift nitrogen load spatially and temporally among losing and gaining stream sections; during low baseflow, sections of the stream resemble nearby springs.
- Arbitrary fixed monitoring locations in carbonate watersheds without synoptic sampling is problematic for assessing management progress.
- Karst areas may reduce nitrogen over short time periods but inputs, conservation effectiveness, and legacy nutrients confound progress.

in Fishing Creek watershed consists of rounded hills and a valley with elevations that range from 300 to 656 m above sea level (U.S. Geological Survey, 2009). Land use within the watershed is predominately undeveloped (71.8%), occurring mostly in the headwaters. The remaining land use consists of cropland (10.6%), pasture (11.5%), and developed areas (6.1%), located primarily in the valley (Figure 1) (U.S. Geological Survey, 2019). Wetlands constitute only 0.12% of the basin located in a few isolated areas. The zonal histogram tool within ArcGIS Pro was used to determine land use and geology percentages (Esri, 2016; Horton et al., 2017; U.S. Geological Survey, 2019). A major interstate transportation corridor passes through the northeast corner of the watershed. Agriculture in the watershed has historically consisted of small dairy farms with hay, corn, and to a lesser extent, small grains such as wheat and rye. In more recent decades, operations have diversified to include poultry, veal, vegetable, and tobacco. Manure applications in the spring are spread on corn and some small grains, while fall spreading of manure is typically applied during pre-planting on wheat and rye acreage, and post-harvest on corn, tobacco, and vegetable crops. Hay fields may receive manure applications in the spring, summer, fall, and sometimes winter. Chemical fertilizer is spread in the spring on corn, vegetables, wheat, and rye. Application of potash is common on such crops as alfalfa hay fields, but the use of lime is limited throughout the watershed (L. Blazure, personal communication, May 6, 2022). Food processing residuals, such as slaughterhouse waste, are permitted to be applied on agricultural fields at various times of the year under specific conditions (Commonwealth of Pennsylvania, 1992; Brandt & Martin, 2001). Point sources within the watershed include a concentrated

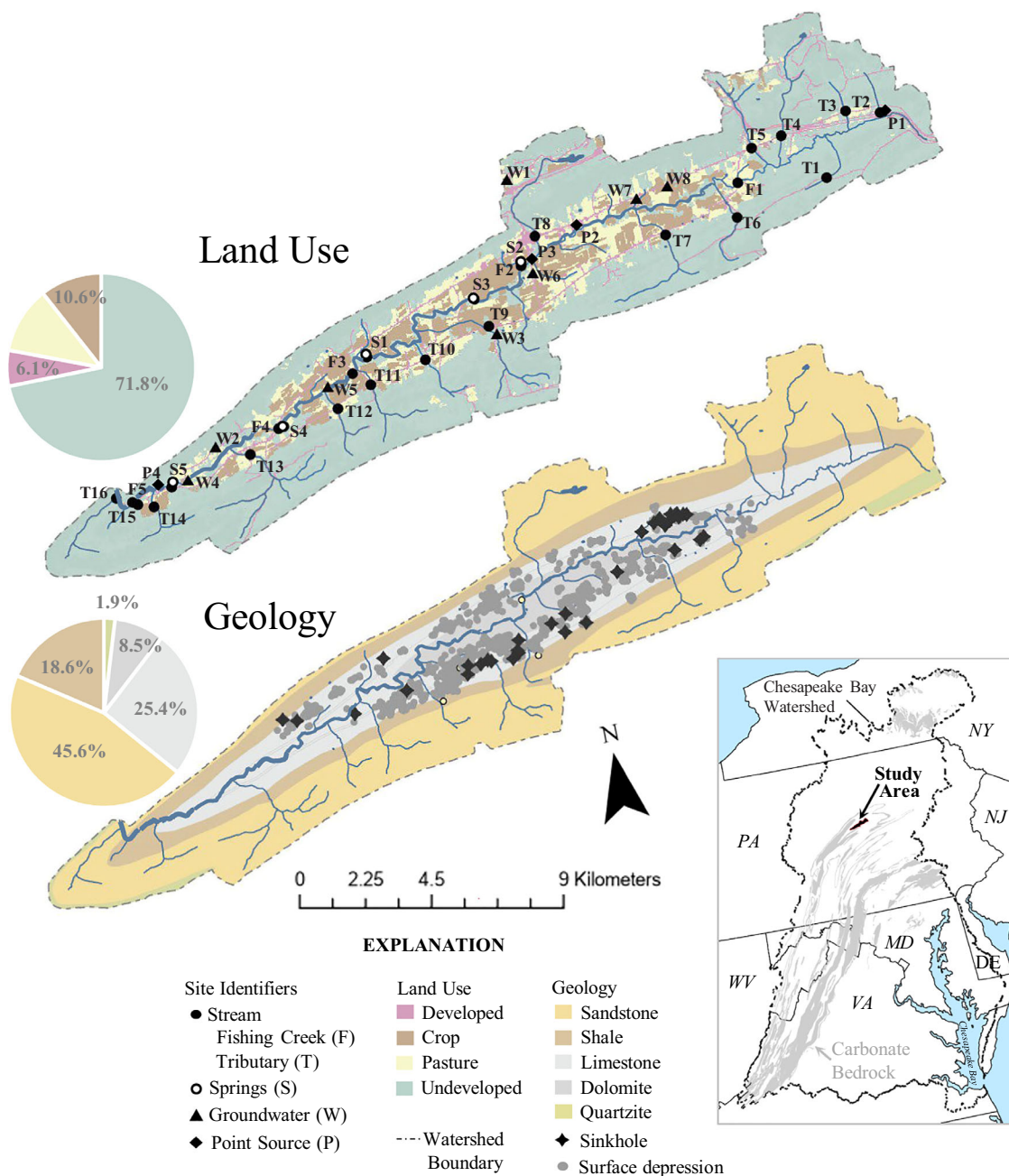


FIGURE 1 Location of study area in the carbonate setting (Brakebill & Kelly, 2000) of the Chesapeake Bay watershed, and sampling sites, land use (USGS, 2019), geology (Horton et al., 2017), and karst features (PA DCNR, 2016) in the Fishing Creek watershed, central Pennsylvania.

animal feeding operation, fish hatchery, meat processing plant, and two wastewater facilities (Table S1; Figure 1).

2.2 | Surface water and groundwater sampling and analysis

Synoptic sampling was used to provide near simultaneous water quality measurements across multiple stream locations in the Fishing Creek watershed. Synoptic sampling was completed in 2019 during high and low base-flow condi-

tions at 26 sites, which included 14 intermittent tributaries and 12 perennial (seven streams and five springs) sampling locations (Table S1). Sampling was performed outside of stormflow influence and during baseflow conditions when the component of streamflow can be attributed to groundwater discharge to streams. Based on data from a nearby stream-gage (U.S. Geological Survey [USGS] station no. 01555000), streamflow from April to August of 2019 was 115% and 82% above the annual median, respectively (U.S. Geological Survey, 2022). These hydrologic conditions allowed for seasonal synoptic sampling of the tributaries and sections

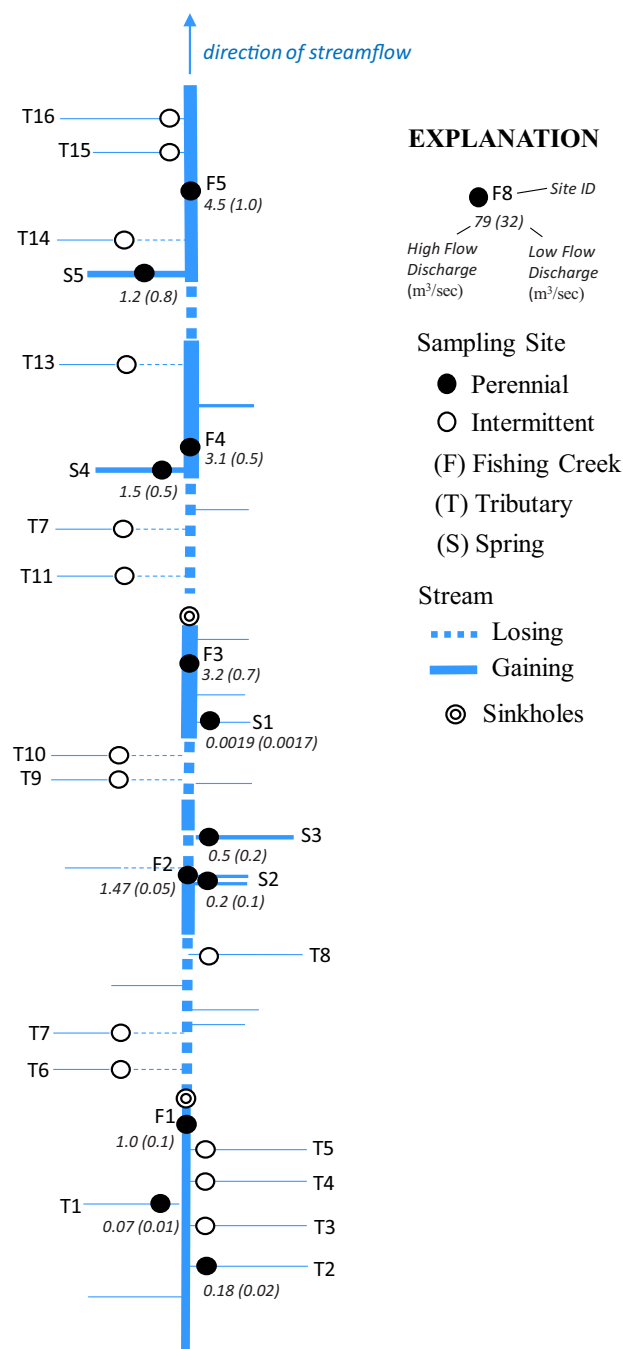


FIGURE 2 Conceptual diagram of surface and subsurface water flow in the Fishing Creek watershed, central Pennsylvania.

of the main Fishing Creek that can be intermittent and lost to the karst during the dry years (Figure 2). Field measurements of water temperature, pH, specific conductance, dissolved oxygen, and nitrate were collected at all the perennial and intermittent sites. The mean of stabilized readings was recorded at six-tenths of the stream depth to represent the average velocity of the water column and provide a homogeneous mixture. At the perennial sites, samples collected for major ions, nutrients, and isotopic chemistry were compos-

ited in a churn from an isokinetic depth-integrating sampler and equal-width-increment sampling (U.S. Geological Survey, n.d.). Grab samples were collected at the centroid of flow for streams that had a velocity of <0.46 m/s. Additionally, streamflow was only measured at perennial sites using volumetric methods or wading with a flow tracker (Turnipseed & Sauer, 2010).

The samples for laboratory measurements of major-ion and nutrient chemistry were filtered through a $0.45\text{-}\mu\text{m}$ pore size disposable capsule filter into a polyethylene bottle, acidified with nitric acid to $\text{pH} < 2$, and analyzed at the USGS National Water Quality Laboratory in Denver, CO. Samples for major ions were analyzed by inductively coupled plasma atomic emission spectrometry, ion chromatography, or automated colorimetry (Fishman, 1993). Nutrient analyses were performed by automated colorimetry with persulfate digestion when required (Fishman, 1993; Patton & Kryskalla, 2011). Samples collected for analysis of nitrogen isotopes were filtered onsite using a sterile Nalgene nitrate-free $0.2\text{-}\mu\text{m}$ filter, preserved with potassium hydroxide until shipped to the USGS Stable Isotope Laboratory in Reston, VA, and analyzed using the bacterial reduction method (Casciotti et al., 2002; Sigman et al., 2001). The nitrogen isotopic delta ($\delta^{15}\text{N}$) values obtained represent the isotopic ratio of $^{15}\text{N}/^{14}\text{N}$ of the sample relative to the standard of atmospheric N_2 , whereas $\delta^{18}\text{O}$ values represent the isotopic ratio of $^{18}\text{O}/^{16}\text{O}$ of the sample relative to the standard of Vienna Standard Mean Ocean Water. The isotopic ratios in this study are reported in parts per thousand (‰), normalized to reference values given in Böhlke et al. (2003), with analytical uncertainties (1-sigma) of approximately $\pm 0.25\text{‰}$ for $\delta^{15}\text{N}$ and $\pm 0.5\text{‰}$ for $\delta^{18}\text{O}$ (Révész & Casciotti, 2007). The $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ isotopic signatures help to indicate if nitrate originated from natural soils, fertilizer, manure, wastewater, or septic areas.

The quality assurance of surface water sampling included both replicate and blank samples. There was no significant variability or bias apparent in the sampling or laboratory techniques for the major ions, nutrients, and nitrogen isotopes. Various graphical and statistical techniques including trilinear diagrams, correlations matrices, and linear correlations were generated for the chemical and isotopic analysis of nitrate in the Fishing Creek watershed.

Historical atmospheric age tracing data were used along with computed saturation indices to infer residence time of the seasonal synoptic sampling events (Focazio et al., 1998; Lindsey et al., 2003; McCleskey et al., 2012). Aqueous speciation with PHREEQC (Parkhurst & Appelo, 2013) was used to compute the charge balance, saturation indices, and relative contributions of ionic conductivity to the specific conductance (Clune & Cravotta, 2019; McCleskey et al., 2012). Charge imbalances were within 5%, and measured values of specific conductance were consistent with corresponding values computed as the sum of ionic conductivities.

Nutrient loads in this study were calculated by multiplying the instantaneous streamflow and concentration results from the lab. In addition to the data collected for this paper, groundwater samples collected during June of 2017 within the range of previous estimates for the residence time of groundwater (Focazio et al., 1998; Lindsey et al., 2003) were also considered from eight domestic wells (median depth of 77.7 m below land surface) sampled for bacteria, major ion, and nutrient chemistry (Clune & Cravotta, 2020) to compare with spring and surface water baseflow samples.

3 | RESULTS AND DISCUSSION

3.1 | Occurrence and distribution of nitrate

The range in nitrate concentrations in springs and streams (<0.04 – 4.99 mg/L), and groundwater (<0.04 – 49.3 mg/L) along with other nutrients and major ions (Table S2) were consistent with previous sampling in Fishing Creek and other carbonate valleys within the Chesapeake Bay watershed (CDS, 2004; Hyer et al., 2016). The concentrations were generally less in tributaries draining the valley sides compared to perennial springs and the main stem of Fishing Creek along the valley. Nitrate concentrations higher than the background concentration for Pennsylvania (0.42 mg/L, Clune et al., 2020) were present in 46% of spring and stream samples in the Fishing Creek watershed.

Nitrate concentrations are low (<1 mg/L) in the forested headwater tributaries within siliciclastic rock types (Figure 1 sandstone and shale), but gradually become elevated along the springs and the main stem of Fishing Creek underlain by the carbonate valley that is influenced by point and non-point inputs of nitrogen (Figure 3a). Although not statistically significant, there was strong evidence (p -value = 0.056 for Wilcoxon rank sum test) that nitrate concentrations were greater during low (summer) baseflow compared to high (spring) baseflow conditions (Figure 4d). During low baseflow, the nitrate concentrations and nitrogen loads in sections of mainstem Fishing Creek resembled those of nearby springs (e.g., F2 and S2; F4 and S4), highlighting their influence on sustaining nitrate concentrations in the surface channel (Figure 3). Further, the change in nitrogen load from low baseflow (spring) to high baseflow (summer) in the main stem of Fishing Creek (sites F1–F5) was sensitive ($>57\%$ absolute percent change) to seasonal hydrologic conditions, whereas loading from springs was relatively consistent ($<56\%$ absolute percent change), which suggests a relatively constant discharge of groundwater of consistent quality (Figure 3b,c). At many sites, the streamflow and associated nitrogen load did not increase in proportion with the corresponding drainage area. For example, during high baseflow, the nitrogen load

was relatively unchanged from F3 to F4, but during low baseflow there was a substantial loss in load downstream between the two sites (Figure 3b,c). Similarly, the discharge and nitrogen load from most springs were not proportional relative to the corresponding topographical derived drainage area and suggests the presence of a conduit network that eventually discharges groundwater from several large perennial springs along Fishing Creek. Complex networks of subsurface flow paths in carbonate settings create challenges for resource managers attempting to mediate the effects of point and non-point sources of nitrate and other contaminants.

During high baseflow in April, the combined nitrogen load computed as the sum of values for F4 plus S5 (811.4 kg/day) was equivalent (within 0.7%) to the downstream site F5 (815.1 kg/day) (Figure 3b). In April, during increased recharge conditions and higher measured load, groundwater discharging into the stream network is likely preventing the reach from losing nitrogen to the subsurface. In contrast, during low summer baseflow in August the load and streamflow at F5 (347.1 kg/day) was approximately 18% less than the sum of the upstream sites F4 plus S5 (409.2 kg/day), indicating substantial streamflow loss downstream of F4 into the subsurface (Figure 3c). S5 had a seasonally uncharacteristic higher nitrogen load during low baseflow conditions (compared to high baseflow load) contributing 72% of nitrogen load to the outlet (F5) of Fishing Creek. S1 is representative of a small spring that has considerable (4.38 mg/L) nitrate concentration with small discharge, but the cumulative effects of similar abundant springs could be significant to overall nitrogen load to the watershed (Figure 3). Overall, spring discharges have higher concentrations during low baseflow compared to high baseflow and may continue to be a source of nitrate to surface waters even during dry periods. Future research outside of baseflow conditions, such as targeting storm events could help determine the magnitude, recharge areas, and pathways of nutrient loading of principal spring discharges (Berglund et al., 2018).

3.2 | Biogeochemical processes and residence time

The carbonate aquifer of the Fishing Creek watershed is susceptible to nitrate contamination due to a combination of thin or well-drained soils and karst features, such as sinkholes and closed surface depressions (Figure 1). Additionally, the anthropogenic inputs of nitrogen coupled with short residence time and oxic conditions ($DO > 0.5$ mg/L) reduces the possibility of any substantial denitrification or attenuation of nitrate in the carbonate aquifer (Tesoriero et al., 2015), most notably shown by the similarities between groundwater and low baseflow conditions during the summer compared to high baseflow (Figure 4b,d). Well depths in meters below

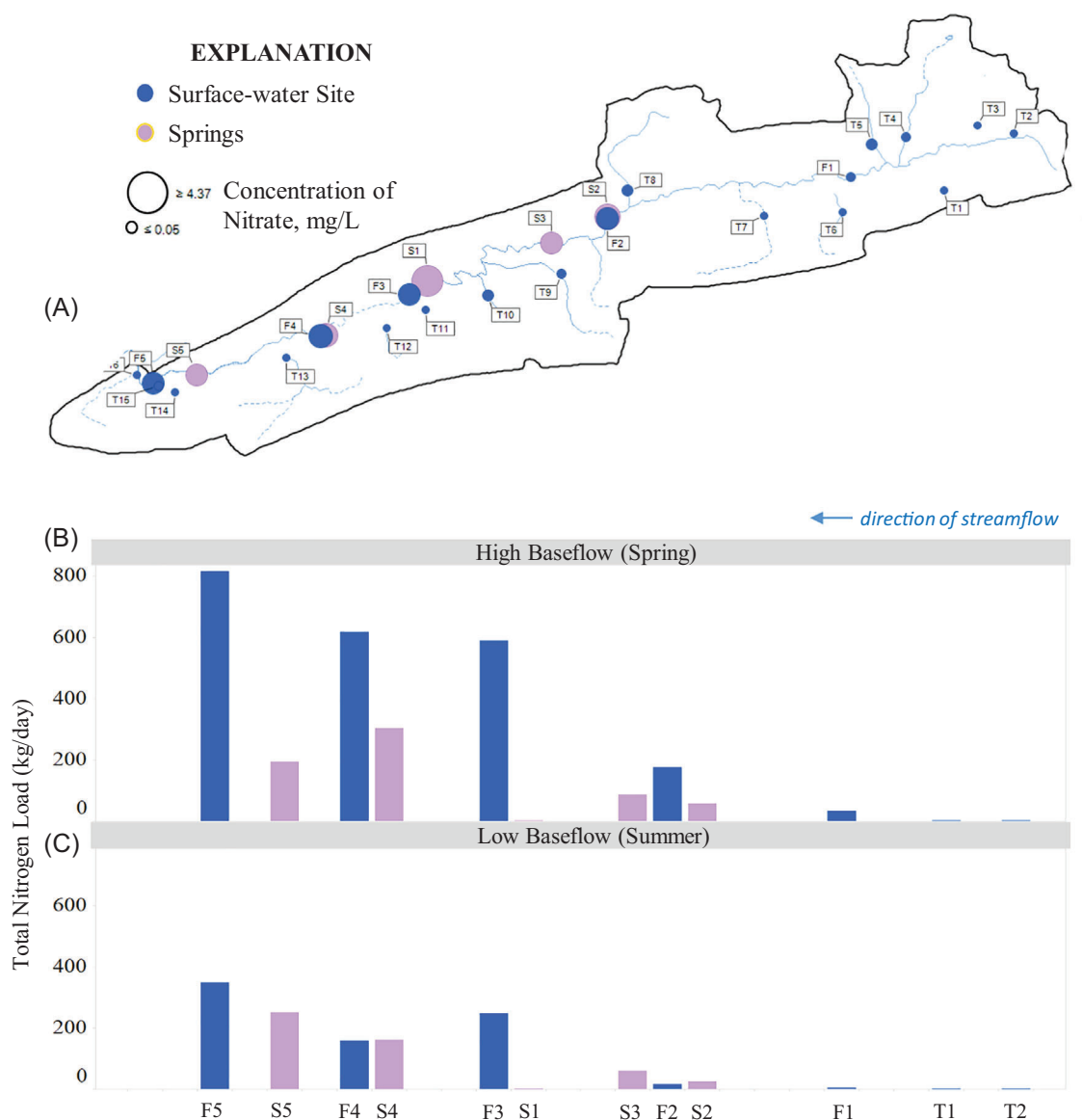


FIGURE 3 Spatial distribution of (A) average nitrate concentrations from instantaneous sensor measurements, and total nitrogen loads during (B) high and (C) low baseflow in 2019 in the Fishing Creek watershed, central Pennsylvania.

ground surface (mbgs) for the deepest wells sampled, CN 502 (121.9 mbgs) and CN 496 (97.5 mbgs) had oxid DO values of 7.94 and 5.57 mg/L, respectively. As a result, land use inputs from non-point and point sources often elevate concentrations of nutrients and major ions in streams compared to natural conditions (Clune et al., 2020).

Water draining the siliciclastic rock along the valley sides has slightly acidic pH that increases when mixed with the Ca-Mg-HCO₃ water type found in the groundwater and springs of the karst valley. Overall pH in the valley was neutral, likely the result of the (1) dissolution of carbonate rock which buffers the effects of acid rain from headwater streams and (2) nitrification of ammonia and organic nitrogen from fertilizer and manure. The calcareous soils remain fertile because the acidic impacts of fertilization are neutralized by weathered dolomite

and limestone deposits. The weathered dolomite and limestone deposits are reflected in the groundwater, springs, and surface water samples by the increased calcium and magnesium concentrations and their relationship with increasing nitrate concentrations (Figure 5a). Furthermore, specific conductance was a good predictor of nitrate because the overall ionic strength is dominated by ions that are commonly related to agricultural applications of fertilizer and manure and the associated weathering of calcareous soils (Figure 5b).

Streams and springs in carbonate settings may include contributions of water and solutes from older groundwater in the aquifer plus relatively recent recharge from streambed leakage and sinkholes that flow through a conduit network (Cravotta, 1988; Jacobson & Langmuir, 1970; Shuster & White, 1971). Interaction with carbonate minerals within

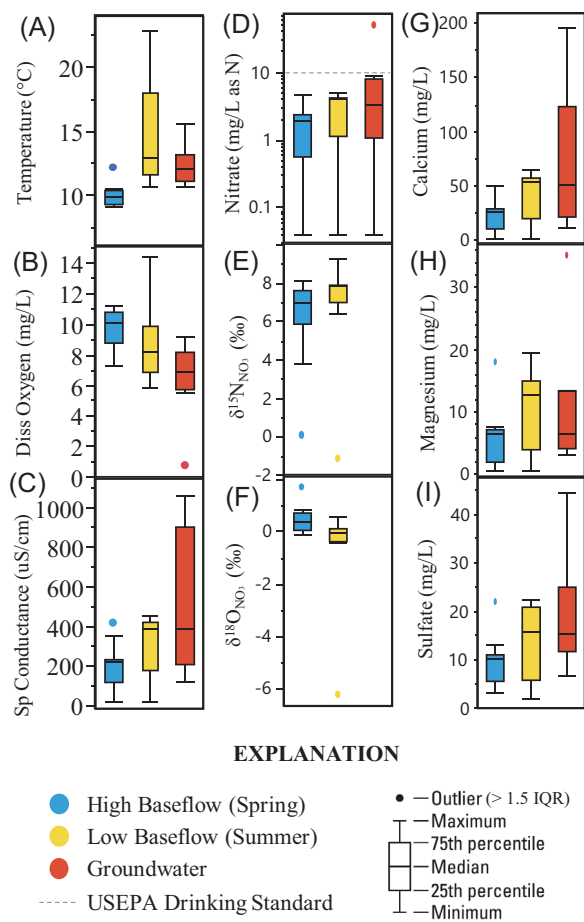


FIGURE 4 Distribution of selected constituents for samples collected at perennial sites in Fishing Creek watershed, central Pennsylvania.

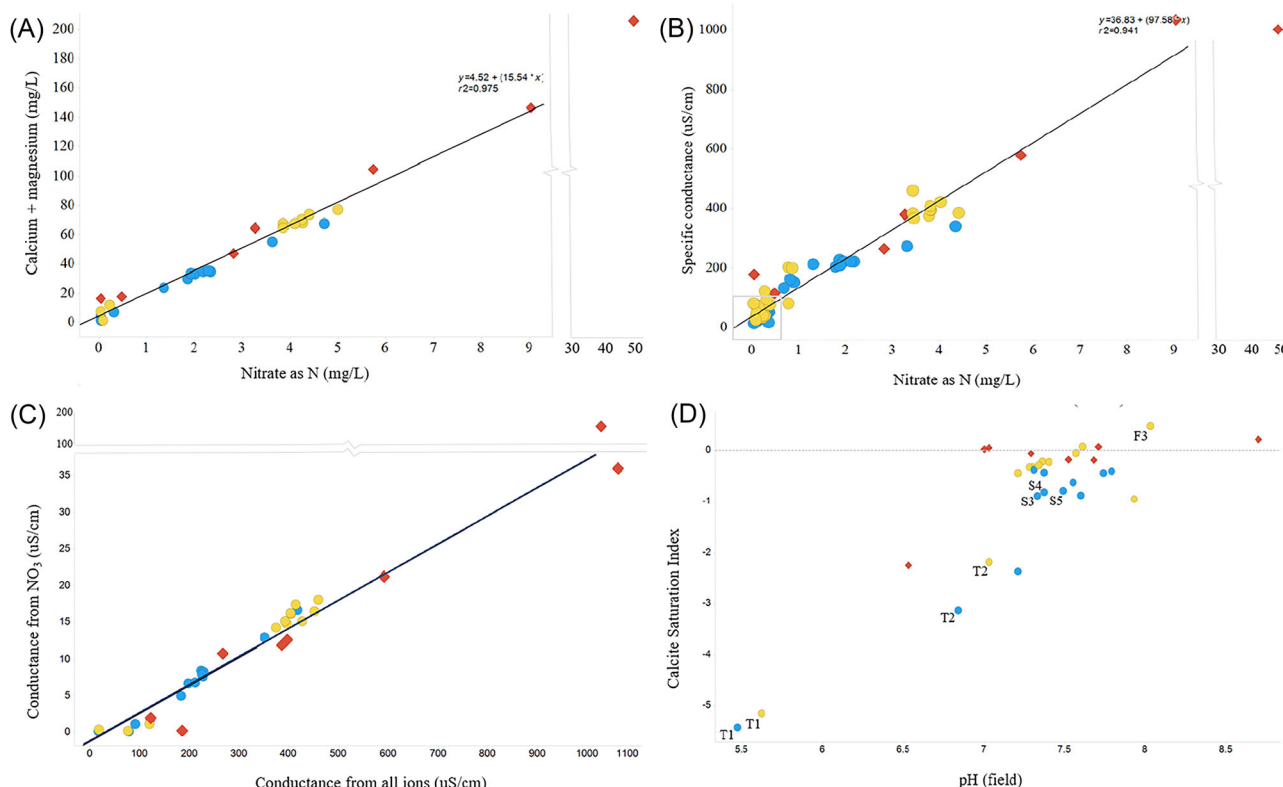
the conduits may be limited by rapid flow rates combined with armoring by silicate and oxide minerals (e.g., Sasowsky & White, 1993). Calcite saturation indices (SI) closer to equilibrium ($0.5 \leq SI \leq 0.5$) are suggestive of a greater fraction of groundwater from the aquifer that has longer residence times as shown in the low baseflow samples from springs and groundwater within the Fishing Creek watershed (Figure 5d). Lower pH and SI values at the same sites during high baseflow conditions reflect a greater fraction of recent recharge. The three largest springs in the valley (S3–S5) sampled during spring high baseflow have more negative SI values, consistent with greater contributions of more acidic, lower pH recharge via sinkholes along the forested valley sides (siliciclastic rock) and corresponding shorter residence times. Springs and surface waters sampled in the low baseflow conditions have a greater fraction of groundwater, similar to the well-water samples from the carbonate aquifer. In general, more extensive mineral dissolution and higher dissolved solids concentrations tend to be associated with deeper and older water systems that have longer flow paths and contact times with surrounding geology (Focazio et al., 1998;

Tesoriero et al., 2005). Based on previous tracer analyses of chlorofluorocarbons (CFCs) at the spring (S4) near the outlet of the watershed, the apparent age of water for the Fishing Creek watershed is only 5–7 years at low baseflow, one of the most rapid residence times collected from springs across the Chesapeake Bay watershed (Focazio et al., 1998; Lindsey et al., 2003) and high baseflow with more recent recharge could have even quicker residence times. Such information on residence time can help resource managers better understand the lag time between implementation of management practices and the water quality response in streams.

The median concentrations of major ions were similar during low baseflow and groundwater samples, but lower during high baseflow in the Fishing Creek watershed (Figure 4; Table S2). Major ion chemistry during low flow is influenced by the greater contribution of older groundwater, whereas high baseflow conditions have a larger portion of younger groundwater recently recharged by precipitation, particularly from small-order forested headwater streams with better water quality or new recharge to sinkholes (Lindsey et al., 2003). Overall, the evolution of water chemistry along the carbonate valley floor of the Fishing Creek aquifer reflects a Ca-Mg-HCO₃ water type (sites F1–F5 and S1–S5 in Figure 6), whereas the siliciclastic rock aquifer along the valley sides has dilute mixed types (sites T1 and T2 in Figure 6). During low baseflow conditions groundwater is the predominant source of spring discharge, as shown by the similar major ion chemistry among the springs and groundwater except for W7 and W8 (discussed in Section 3.3), which are likely strongly influenced by local anthropogenic contaminant sources (Figure 6). Specific conductance increases from the dilute siliciclastic headwater tributaries to the carbonate valley and anthropogenic signatures in water quality can also be observed in some of the chemical and isotopic analysis.

3.3 | Potential sources of nitrate

The vast majority of nitrogen inputs to the Fishing Creek watershed are from a steady application of manure, fertilizer, and declining atmospheric deposition over the past 25 years (Devereux et al., 2022) (Figure 7). The headwater streams (T1 and T2) were representative of water that is predominately precipitation with minimal forest nutrient recycling and had very low nitrate concentration and $\delta^{15}\text{N}$ values consistent with an atmospheric deposition source (Figures 6 and 8) (Garten, 1992; Lindsey et al., 2003; Paerl & Fogel, 1994). The presence of elevated concentrations of sodium and chloride at T2 may be influenced by road deicing chemicals and/or wastewater from an upstream source (Figures 1 and 6a; Table S2). The $\delta^{15}\text{N}$ values from the high and low baseflow samples in the carbonate valley shows a manure and wastewater isotopic signature ($>+6\text{‰}$) or from mixed sources



EXPLANATION

- High Baseflow (Spring)
- Low Baseflow (Summer)
- ◆ Groundwater site

FIGURE 5 Relation of (A) nitrate and calcium plus magnesium, (B) nitrate specific conductance, (C) conductance from nitrate compared to all ions, and (D) calcite saturation with pH for samples collected in Fishing Creek watershed, central Pennsylvania.

(fertilizer, manure, and wastewater) (Figure 8a). For example, the highest $\delta^{15}\text{N}$ values of nitrate reported at co-located sites S2 and F2 during the summer (greater than 8‰) were similar to results reported from previous studies where nitrate was indicative of animal or human waste sources (Hyer et al., 2001; Lindsey et al., 2003; Mayer et al., 2002). Furthermore, the nitrate isotopic signature of fertilizer, manure, and wastewater found in the baseflow of the streams and springs may also be present in groundwater. For example, multiple lines of evidence suggest anthropogenic impacts may be influencing the groundwater water quality of domestic wells W7 and W8 (Figure 1), including (1) elevated nitrate concentrations, specific conductance, and bacteria levels (Table S2); (2) signatures in bromide/chloride ratios (Figure 8b); and (3) nitrate and chloride contributions in major ion chemistry (Figure 6b). Groundwater wells in similar carbonate settings with a high sinkhole density dominated by conduit flow have been shown to have significantly higher nitrate concentrations than wells in low sinkhole-density areas (Lindsey et al., 2010).

Seasonally, $\delta^{15}\text{N}$ values were higher during the summer when low baseflow in streams is primarily older water containing a higher fraction of anthropogenic nitrogen sources than younger groundwater recharged from more recent precipitation (Figures 4e and 8a; Table S2).

Nitrate and chloride ratios suggest agriculture and/or wastewater dominates low baseflow, but a substantial nitrogen load comes from mixed sources during the high baseflow (Figure 8c). Under natural conditions, observed concentrations of nitrate and chloride and associated nitrate/chloride ratios reflect contributions from atmospheric precipitation. Low nitrate and nitrate/chloride ratios are maintained by denitrification and other processes that attenuate nitrate. In contrast, nitrogen contributions from additional sources (manure, fertilizer, wastewater, etc.) result in a relative excess of nitrate compared to chloride and other ions. Although nitrate attenuation may occur to some extent, processes such as denitrification and uptake are insufficient to maintain a balance between nitrate and chloride in the agricultural and/or wastewater environment.

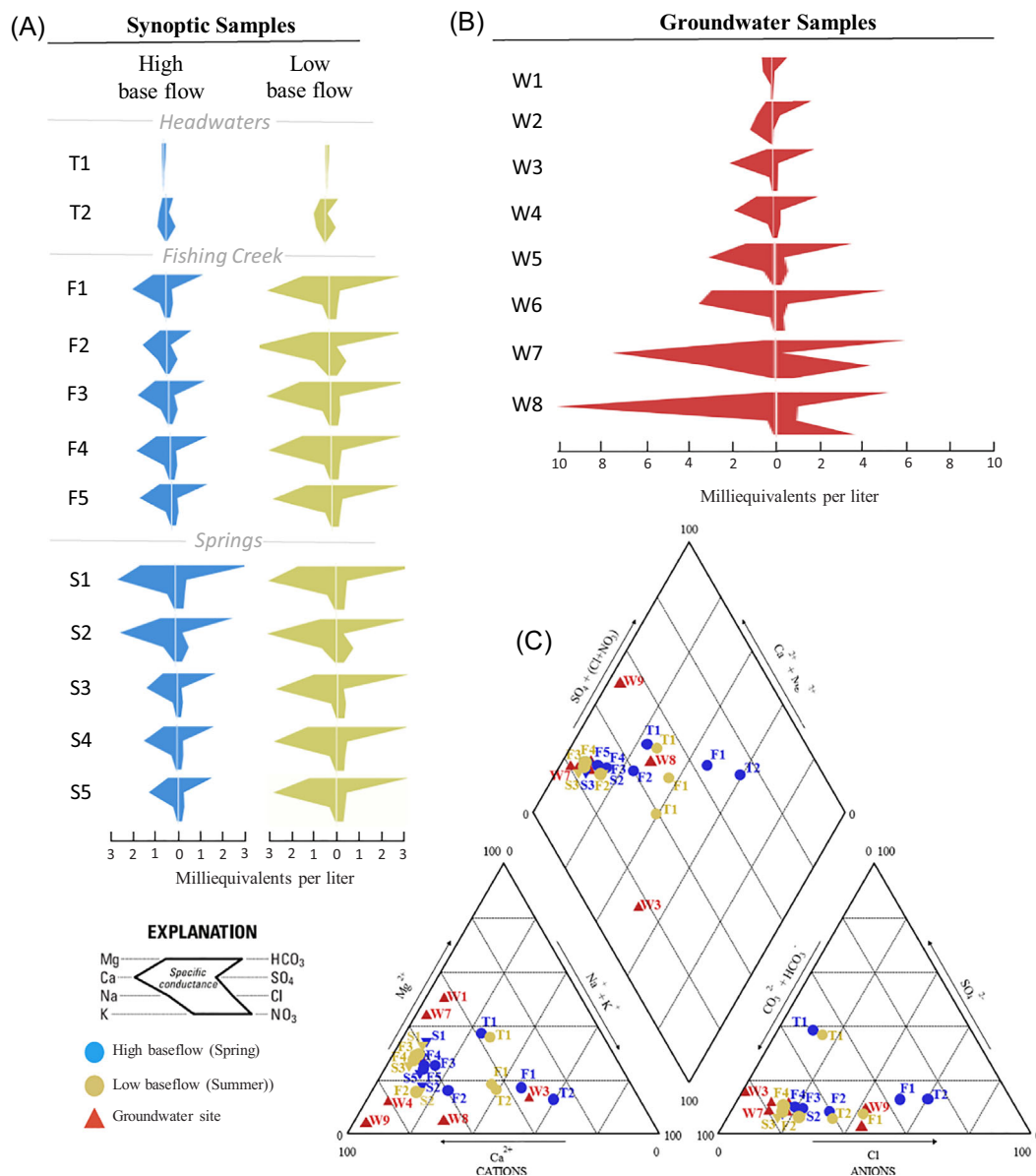


FIGURE 6 Relative abundance patterns of constituents in (A) synoptic and (B) groundwater samples, and (C) a trilinear diagram of major ions within Fishing Creek watershed, central Pennsylvania.

Denitrification in the aquifer of Fishing Creek is unlikely due to the prevalence of oxic conditions in groundwater (Figures 4b and 8c) and median nitrate concentrations are likely to remain stable in groundwater and low stream base-flow conditions as long as sources remain constant (Figure 4d; Table S2). Furthermore, denitrification generally would result in the enrichment of both $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ relative to source materials and be reflected by a positive slope, which is not present in Figure 8a. Relatively constant values for $\delta^{18}\text{O}$ with corresponding increased values for $\delta^{15}\text{N}$ that reflect mixed sources (fertilizer, manure, and wastewater) support the interpretation that denitrification is negligible. Conducting a study along losing stream sections or a flow path from groundwater to streams could help resource managers better

identify the spatial and temporal impact of land use on stream water quality and potential non-conservative denitrification opportunities in karst settings (Husic et al., 2020; Laincz, 2011).

3.4 | Research and management challenges

The complex hydrogeology and land-use characteristics of carbonate watersheds create challenges for research and management efforts intended to restore estuaries such as the Chesapeake Bay. Established fixed-station monitoring programs often utilize existing infrastructure (bridge crossing, existing streamgages, etc.) for ease of access and do not

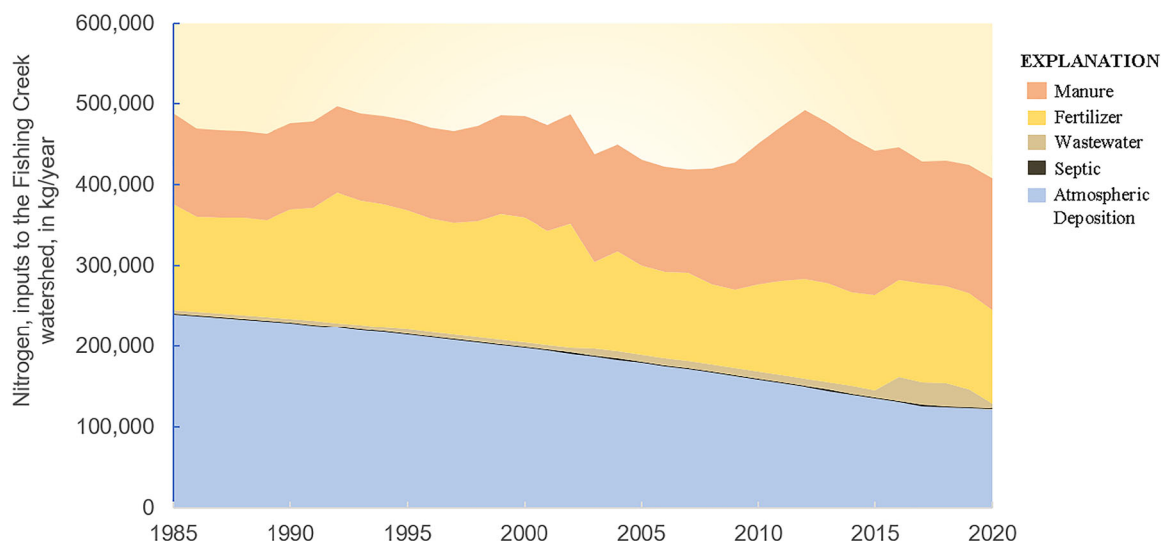


FIGURE 7 Nitrogen inputs from manure, fertilizer, wastewater, septic, and atmospheric sources in Fishing Creek watershed, central Pennsylvania, 1985–2020 (Data Source: Devereux et al., 2022).

perform the watershed-wide synoptic sampling methods outlined in this study (i.e., nitrate sensor, water quality samples, and discharge seepage measurements). Arbitrary selection of fixed monitoring sites along losing or gaining stream sections in carbonate systems, such as the Fishing Creek watershed, without prior characterization of water quality and development of a conceptual stream network could be problematic for accurate assessment of existing conditions, long-term trends, and progress of management actions (Figure 2). For example, Fishing Creek contains two of the largest springs in Pennsylvania, and the discharge and associated nitrogen load are likely connected through the carbonate subsurface. Beyond coupling synoptic sampling with fixed monitoring, future research could include an inventory of karst features and dye tracing, bacteria and wastewater markers, isotope mixing models, baseflow separation, and microbiological indicators (Brahana et al., 2012; Canion et al., 2020; Husic et al., 2020; Katz & Griffin, 2008; Miller et al., 2017; Xue et al., 2012). Additionally, field and geospatial observations of sinkholes in this study could identify the most effective sinkhole buffer measures (similar to riparian buffers) to better protect the aquifer from direct runoff of nutrients. Improved electronic reporting of nitrogen inputs from point sources would help identify sources that may be affecting water quality, especially during low flow. A database of nitrogen inputs from point sources would also help models be more accurate in representing non-traditional point sources (food residual waste, biosolids, hatchery waste, etc.) that are not quantified in regional datasets. Management strategies in the Fishing Creek and similar karst watersheds have the potential to reduce nitrate concentrations within a decadal time scale, but continued nitrogen inputs (Figure 7), effectiveness and placement of conservation practices (i.e., protection of sinkholes), and

legacy storage and episodic release of nutrients and sediments stored within the karst cavities and the valley could confound progress (Fidelibus et al., 2017; Herman et al., 2008).

4 | CONCLUSIONS

These findings indicate field measurements of water quality combined with synoptic sampling over a short period of time can improve understanding of the relative contribution of nitrogen sources in a small carbonate watershed. The complex karst hydrology of Fishing Creek demonstrated that the streamflow and nitrogen load did not increase with drainage area and was often lost to sinkholes that discharged downstream along large springs. Additionally, the observed seasonal hydrologic conditions shift the nitrogen load spatially among losing and gaining stream sections originating from a mixture of manure, fertilizer, and wastewater sources with low potential for denitrification. Effective nitrogen reduction strategies located in carbonate watersheds with complex groundwater/surface water flow interactions and large inputs of nitrogen, short residence times, and minimal denitrification, may offer the best opportunity to provide a rapid response toward achieving water quality goals for the Chesapeake Bay and similar estuaries globally. Foremost, such management actions guided by monitoring could benefit human health as carbonate aquifers constitute a drinking water resource for nearly a quarter of the world's population (Ford & Williams, 2007; Goldscheider et al., 2020). Given the substantial investment in management practices, fixed non-tidal monitoring sites in karst watersheds could incorporate synoptic surface water sampling to ensure properly monitored long-term progress, help inform management actions,

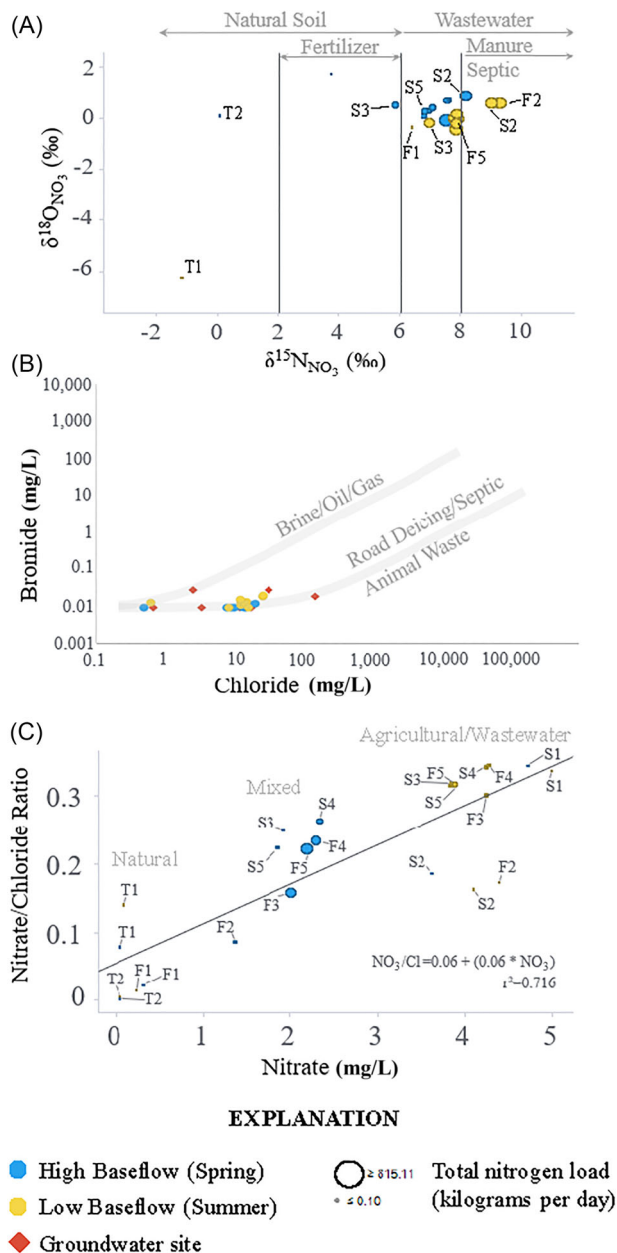


FIGURE 8 Relation of (A) nitrogen and oxygen isotopes of nitrate, (B) chloride and bromide ions, and (C) nitrate and chloride ions for samples collected from Fishing Creek watershed, northcentral Pennsylvania.

and protect aquatic and human health. Future research could include an inventory of karst features and dye tracing, bacteria and wastewater markers, isotope mixing models, baseflow separation, and microbiological indicators to enhance understanding of the local complex hydrology and the sources of nutrients.

AUTHOR CONTRIBUTIONS

John W. Clune: Conceptualization; data curation; formal analysis; investigation; methodology; project administra-

tion; visualization; writing—original draft; writing—review and editing. **Charles A. Cravotta III:** Formal analysis; visualization; writing—review and editing. **Admin Husic:** Writing—review and editing. **Hilary J. Dozier:** Data curation; investigation; methodology. **Kurt E. Schimdt:** Formal analysis; visualization.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Using the station identification numbers in Table S1, all water quality results used for analysis in this study can be retrieved through the National Water Information System (NWIS) with user-defined queries using USGS web services (U.S. Geological Survey, 2022) or with the data Retrieval R package (Hirsch & De Cicco, 2015; Hirsch et al., 2015a, 2015b).

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SUPPORTING INFORMATION

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